Municipal Road Health & Maintenance Brief

Autonomous Surveillance Telemetry & Pothole Hotspot Report • Generated October 16, 2026 at 20:17 UTC

TOTAL DETECTIONS	TIER 1 (CRITICAL)	TIER 2 (MODERATE)	ROAD HEALTH INDEX
33	15	6	30%

Executive Summary & Risk Assessment

During recent mobile dashcam and vision surveillance across the Punjab monitoring sector, the YOLOv8 autonomous pipeline logged **33 road surface anomalies** (mean AI detection confidence: **69.5%**). A total of **18 cavities/potholes (D40)** and **15 structural cracks (D10/D20)** were geolocated. Currently, **15 critical hazards** represent Tier 1 severe road failures exceeding standard depth and area thresholds, posing immediate risk to vehicular safety and tire integrity.

DBSCAN Spatial Hotspots & High-Density Priority Zones

Cluster ID	Center Coordinates	Hazard Count	Radius	Dominant Tier	Municipal Action
Hotspot #1	32.5738° N, 74.0766° E	**8 hazards**	~97 m	**Tier 1**	Emergency Cold Patch (24h)

Critical Hazards Awaiting Maintenance Dispatch (Tier 1 Sample)

Detection ID	Class Name	Confidence	GPS Location	Timestamp (UTC)
8a97ac42	POTHOLE	94.0%	32.5735, 74.0762	2026-10-17 01:00:00
5e990a73	POTHOLE	82.9%	32.5743, 74.0771	2026-10-16 20:12:21
81b8871a	POTHOLE	76.4%	32.5742, 74.0770	2026-10-16 20:12:20
7228bc5c	POTHOLE	90.1%	32.5741, 74.0768	2026-10-16 20:12:20
8fea5e31	POTHOLE	73.5%	32.5739, 74.0767	2026-10-16 20:12:19

Municipal Action Recommendation:
- **Emergency Cold-Patching:** Dispatch rapid-response civil teams to Hotspot #1 immediately.
- **Quarterly Budget Allocation:** Reserve asphalt emulsion supply for high-stress monsoon degradation.
- **Surveillance Cadence:** Maintain bi-weekly dashcam telemetry capture along major logistics corridors.